
Data Analytics Internship - One-Day Assessment

Databricks Machine Learning: Predictive Maintenance Early Warning System

Duration	One day preparation + class explanation
Mode	Databricks notebook using Spark ML and MLflow
Assessment Type	Company-style ML case study. No ready-made solution will be provided.

Important Note on AI Assistance

Students are allowed to use AI tools to research the topic, understand Databricks syntax, explore Spark ML approaches, understand MLflow logging, and clarify machine learning concepts. However, the final notebook, decisions, interpretation, and class explanation must be understood by the student. Copying AI-generated code without understanding it will not be considered a valid submission.

Dataset

Dataset name: AI4I 2020 Predictive Maintenance Dataset

Source: UCI Machine Learning Repository

Dataset page: <https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset>

Direct download: <https://archive.ics.uci.edu/static/public/601/ai4i%2B2020%2Bpredictive%2Bmaintenance%2Bdataset.zip>

The dataset represents an industrial predictive maintenance scenario. It contains machine operating conditions such as air temperature, process temperature, rotational speed, torque, tool wear, product type, and failure indicators. The task is classification-oriented, but students are expected to think beyond simple accuracy and connect the model output to maintenance decisions.

Business Problem

A manufacturing company runs a set of automated machines on a production line. Unexpected machine failure stops production, delays customer delivery, increases repair cost, and affects plant-level productivity. The maintenance team does not want a dashboard that only shows historical failures. They want an early warning system that can score each machine reading and identify whether the machine is likely to fail soon.

The company also has a practical constraint. Maintenance engineers cannot inspect every machine reading. They need a model that helps them prioritize which machines require urgent attention, which machines should be watched, and which machines can continue normal operation.

Your role is to build a Databricks-based machine learning workflow that converts the raw dataset into an ML-ready table, trains a failure-risk model using Spark ML, tracks the experiment using MLflow, and produces a maintenance decision output that can be explained to both technical and business users.

Assessment Requirement

Build a Databricks notebook for this case. The notebook should show how a company analyst or junior data scientist would approach predictive maintenance using Spark ML inside Databricks. The work should not stop at model training. The final result should connect model predictions to maintenance action.

Students are expected to make their own decisions about feature preparation, leakage prevention, model selection, evaluation metrics, and risk-tier design. Any reasonable approach is accepted if it is technically correct, business-aware, and clearly explained in class.

Core Challenge

The model should not merely predict whether machine failure happens. It should support a practical maintenance decision. Create a final output where each machine reading is assigned a failure risk score and a maintenance action tier. For example, the output may classify readings into categories such as Normal, Watch, and Urgent, based on the probability or risk score created by the model.

Students should also think carefully about columns that may behave like post-failure explanation fields. If a column directly explains the failure after it has happened, it should not be blindly used as an input feature. This is part of the assessment: students must identify possible leakage risks and justify their feature choices.

What the Notebook Should Demonstrate

The notebook should demonstrate a complete Databricks ML usage flow. It should load the dataset, create an ML-ready version of the data, apply feature engineering suitable for Spark ML, train at least one classification model, evaluate the model, log results using MLflow, and save useful outputs as Delta tables where appropriate.

The work should cover the ideas discussed in class: prediction problem framing, leakage-aware feature selection, categorical handling, numeric feature preparation, Spark ML pipeline thinking, model evaluation, experiment tracking, and business interpretation. Students may research the required implementation details using AI, documentation, or other learning resources.

Expected Final Outcome

At the end of the assignment, each student or team should be able to explain the following in class:

- What exact business decision the model supports.
- How the target variable was used and what leakage risks were considered.
- Which features were used, which were excluded, and why.
- How Spark ML was used to build the model workflow.
- Which evaluation metrics were chosen and why those metrics matter for maintenance decisions.
- How MLflow was used to track model runs, parameters, and metrics.
- How the final risk-tier output can help the maintenance team act before failure becomes costly.
- What changes would be needed before this becomes a production-ready system.

Business Evaluation Angle

Do not judge the model only by accuracy. In predictive maintenance, the cost of missing a true failure may be much higher than the cost of raising a false warning. Students should explain how they think about false negatives, false positives, recall, precision, and practical maintenance capacity. A technically acceptable model is not enough; it must make sense for the factory operation.

Class Explanation Format

Each student or team will explain their work in class. The explanation should sound like a company project discussion, not a textbook definition. Explain the business problem, the ML design choices, the model result, the MLflow tracking, and the final maintenance recommendation output. The expectation is that students should be able to defend their decisions if asked follow-up questions.

Submission Guidance

Submit the Databricks notebook and any supporting output screenshots or table names needed to verify the work. The main scoring focus will be on the thinking, implementation quality, business interpretation, and ability to explain the notebook. A simple model with correct leakage handling and strong explanation is better than an advanced model with unclear business logic.

Interview Relevance

This assessment is designed to help students answer interview-style questions such as: How do you use Databricks for machine learning? How do you prepare an ML-ready dataset? How do you avoid leakage? How do you handle imbalanced classification? How do you use MLflow? How do you convert model predictions into business action? These questions are more valuable than simply saying that a model was trained.

Final Reminder

The assignment is not about finding a perfect model. It is about learning how to build and explain a company-style ML workflow inside Databricks. Use AI assistance if needed, but make sure you understand every major decision well enough to explain it clearly.